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1 Energy in Thermal Physics

1.1 Thermal Equilibrium

This section introduces the concept of temperature.

- **Thermal Equilibrium:** When two objects are in thermal contact, they spontaneously exchange energy (as **heat**) until they reach a state of equilibrium[p. 65].
- **Temperature (T):** Temperature is defined as the quantity that is the same for both objects when they are in thermal equilibrium[p. 62].
- **Absolute Zero:** The zero point of the absolute temperature scale (Kelvin, K) is defined as $-273.15^\circ C$ [p. 79]. All calculations in thermodynamics should use an absolute scale like Kelvin.

1.2 The Ideal Gas

This section relates temperature to the microscopic properties of an ideal gas.

- **Ideal Gas Law:** The macroscopic equation of state for n moles of an ideal gas[p. 103].

$$PV = nRT \quad (1.1)$$

Here R is the universal gas constant, $R = 8.315 \text{ J/mol} \cdot K$ [p. 107].

- In terms of the number of molecules N , the law is[p. 113]:

$$PV = NkT \quad (1.5)$$

Where k is Boltzmann's constant, $k = 1.381 \times 10^{-23} \text{ J/K}$ [p. 115].

- **Average Kinetic Energy:** From the microscopic model, the average translational kinetic energy of a single gas molecule is directly proportional to temperature [p. 162-164]. This is a key result linking the microscopic and macroscopic views.

$$\overline{K}_{\text{trans}} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}kT \quad (1.17)$$

- **RMS Speed:** The root-mean-square (rms) speed of gas molecules depends on their mass m and the temperature T [p. 171].

$$v_{\text{rms}} \equiv \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}} \quad (1.21)$$

1.3 Equipartition of Energy

This section generalizes the energy-temperature relation.

- **Degree of Freedom (f):** A "degree of freedom" is any way a molecule can store energy, where the formula for the energy is quadratic in a coordinate or momentum (e.g., $\frac{1}{2}mv_x^2$, $\frac{1}{2}I\omega_x^2$, $\frac{1}{2}k_s x^2$) [p. 181, 183].
- **Equipartition Theorem:** At temperature T , the average energy of any single quadratic degree of freedom is $\frac{1}{2}kT$ [p. 185].
- **Total Thermal Energy (U_{thermal}):** For a system of N molecules, each with f "active" (not "frozen out") degrees of freedom [p. 185]:

$$U_{\text{thermal}} = N \cdot f \cdot \frac{1}{2}kT \quad (1.23)$$

- Monatomic gas: $f = 3$ (translation only) [p. 188].
- Diatomic gas (room temp): $f = 5$ (3 translation + 2 rotation) [p. 188, 191].
- Solid: $f = 6$ (3 translation K.E. + 3 potential E.) [p. 192].

1.4 Heat and Work

This section defines the First Law of Thermodynamics.

- **Internal Energy (U):** The total energy contained within a system [p. 205].
- **Heat (Q):** A spontaneous transfer of energy from one object to another due to a temperature difference [p. 203].
- **Work (W):** Any other transfer of energy into or out of a system [p. 204].
- **First Law of Thermodynamics:** Energy is conserved. The change in a system's internal energy (ΔU) is the sum of heat added *to* the system (Q) and work done *on* the system (W) [p. 206-207].

$$\Delta U = Q + W \quad (1.24)$$

1.5 Compression Work

This section details the most common form of work in thermodynamics.

- **Quasistatic Work:** For a slow, reversible compression or expansion of a gas, the work done *on* the system is [p. 227-231, 241]:

$$W = - \int_{V_i}^{V_f} P(V) dV \quad (1.29)$$

This is the negative of the area under the P - V curve [p. 240].

- **Isothermal Process:** If T is constant, for an ideal gas $P = NkT/V$ [p. 263-264]. The work done is:

$$W = -NkT \int_{V_i}^{V_f} \frac{1}{V} dV = NkT \ln \left(\frac{V_i}{V_f} \right) \quad (1.30)$$

- **Adiabatic Process:** A process with no heat transfer ($Q = 0$)[p. 271]. From the First Law, $\Delta U = W$ [p. 272].
- For a quasistatic adiabatic process on an ideal gas, the following relations hold:

$$VT^{f/2} = \text{constant} \quad (1.39)$$

$$V^\gamma P = \text{constant} \quad (1.40)$$

Where $\gamma = (f + 2)/f$ is the **adiabatic exponent**[p. 290].

1.6 Heat Capacities

This section defines quantities that measure how a system's temperature changes when energy is added.

- **Heat Capacity (C):** The heat required to raise the temperature, per degree increase[p. 305].

$$C \equiv \frac{Q}{\Delta T} \quad (1.41)$$

- **Specific Heat Capacity (c):** Heat capacity per unit mass, $c = C/m$ [p. 306-307].
- **Heat Capacity at Constant Volume (C_V):** No PV -work is done ($W = 0$)[p. 310].

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V \quad (1.44)$$

From equipartition ($U = \frac{f}{2}NkT$):

$$C_V = \frac{Nfk}{2} \quad (1.46)$$

- **Heat Capacity at Constant Pressure (C_P):** The system does work as it expands[p. 314].

$$C_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P \quad (1.45)$$

For an ideal gas, this simplifies to a crucial relation[p. 337]:

$$C_P = C_V + Nk \quad (1.48)$$

- **Latent Heat (L):** The heat required to completely change the phase of a substance (e.g., melt or boil) at constant T [p. 355-357].

$$L \equiv \frac{Q}{m} \quad (1.50)$$

- **Enthalpy (H):** A useful thermodynamic potential for constant-pressure processes[p. 362].

$$H \equiv U + PV \quad (1.51)$$

For a quasistatic process at constant P where the only work is PV -work, the change in enthalpy is exactly equal to the heat added [p. 371-375]:

$$\Delta H = Q \quad (\text{from Eq. 1.55})$$

This provides an alternative (and often easier) way to calculate C_P [p. 376]:

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P \quad (1.56)$$

2 The Second Law

2.1 Two-State Systems

This section introduces statistical concepts using simple systems.

- **Microstate:** The specific state of each individual particle in a system (e.g., HHT for three coins)[p. 500].
- **Macrostate:** A generalized description of the system (e.g., "2 heads")[p. 500].
- **Multiplicity (Ω):** The number of microstates corresponding to a given macrostate[p. 500].
- For N coins, the multiplicity of n heads is given by the binomial coefficient:

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{n!(N-n)!} \quad (2.6)$$

- **Two-State Paramagnet:** A model system of N magnetic dipoles, each of which can be "up" or "down"[p. 525-526]. This is mathematically identical to the coin-flipping system[p. 531].

2.2 The Einstein Model of a Solid

This section models a solid as a collection of quantum harmonic oscillators.

- **Einstein Solid:** A model of a solid as N independent harmonic oscillators, all with the same frequency f [p. 540].
- The energy of each oscillator is quantized in discrete units of $\epsilon = hf$ [p. 533].
- The multiplicity of an Einstein solid with N oscillators and q energy units is[p. 546]:

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!(N - 1)!} \quad (2.9)$$

2.3 Interacting Systems

This section examines two systems (e.g., two Einstein solids) that can exchange energy.

- The total multiplicity Ω_{total} of a combined macrostate (e.g., q_A and q_B) is the product of the individual multiplicities: $\Omega_{\text{total}} = \Omega_A \Omega_B$ [p. 560].
- **Fundamental Assumption of Statistical Mechanics:** In an isolated system in thermal equilibrium, all accessible microstates are equally probable [p. 587].
- This implies that the system will most likely be found in the macrostate with the **greatest multiplicity** [p. 591, 618].
- **Second Law of Thermodynamics (Statistical Version):** Multiplicity tends to increase [p. 618].

2.4 Large Systems

This section applies mathematical tools to systems with large numbers ($N \sim 10^{23}$).

- **Stirling's Approximation:** For large N , $N!$ is unwieldy. We use the approximation:

$$\ln N! \approx N \ln N - N \quad (2.16)$$

A more accurate version is $N! \approx N^N e^{-N} \sqrt{2\pi N}$ [p. 645].

- **Multiplicity of a Large Einstein Solid:** In the high-temperature limit ($q \gg N$), the multiplicity is approximately [p. 665]:

$$\Omega(N, q) \approx \left(\frac{eq}{N}\right)^N \quad (2.21)$$

- **Sharpness of the Multiplicity Function:** For two interacting large systems, the Ω_{total} function is an extremely sharp **Gaussian** peak centered on the most probable macrostate (e.g., where energy is shared equally) [p. 670, 681].
- **Thermodynamic Limit:** For macroscopic systems, fluctuations away from the most likely macrostate are unmeasurably small. We can assume the system *is* in its most likely macrostate [p. 689].

2.5 The Ideal Gas

This section derives the multiplicity of a monatomic ideal gas.

- **Multiplicity of a Monatomic Ideal Gas:** The derivation involves counting states in a $3N$ -dimensional "momentum space." The final result depends on V and U as:

$$\Omega(U, V, N) = f(N) V^N U^{3N/2} \quad (2.41)$$

where $f(N)$ is a complicated function of N [p. 743].

- **Volume Dependence:** $\Omega \propto V^N$. The probability of finding N molecules in a sub-volume $V/2$ is $(1/2)^N$, which is astronomically small for large N [p. 762].

2.6 Entropy

This section defines entropy, the central concept of the Second Law.

- **Entropy (S):** Entropy is defined as the logarithm of the multiplicity[p. 768].

$$S \equiv k \ln \Omega \quad (2.45)$$

Entropy is a measure of the number of available microstates.

- **Second Law of Thermodynamics (Entropy Version):** Any large system in equilibrium will be found in the macrostate with the greatest entropy. Entropy tends to increase [p. 779-780].
- **Entropy of an Ideal Gas (Sackur-Tetrode Equation):**

$$S = Nk \left[\ln \left(\frac{V}{N} \left(\frac{4\pi mU}{3Nh^2} \right)^{3/2} \right) + \frac{5}{2} \right] \quad (2.49)$$

- **Free Expansion:** An adiabatic expansion into a vacuum. $Q = 0$ and $W = 0$, so $\Delta U = 0$ [p. 794]. The entropy increases because the volume increases[p. 795, 792]:

$$\Delta S = Nk \ln \left(\frac{V_f}{V_i} \right) \quad (2.51)$$

- **Entropy of Mixing:** When two different gases mix, the total entropy increases. For two identical gases, it does not[p. 802, 807]. This is resolved by the fact that identical particles are indistinguishable[p. 816].
- **Irreversible Processes:** Processes that create new entropy (increase Ω_{total}) and cannot happen in reverse[p. 822].

3 Interactions and Implications

3.1 Temperature

This section provides the fundamental, statistical definition of temperature.

- When two systems (A and B) are in thermal equilibrium, their total entropy ($S_{\text{total}} = S_A + S_B$) is maximized.
- At equilibrium, the slopes of their entropy-energy graphs are equal:

$$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B} \quad (3.3)$$

- **Temperature (T):** This common quantity is defined as the reciprocal of the slope of the entropy-energy graph.

$$\frac{1}{T} \equiv \left(\frac{\partial S}{\partial U} \right)_{N,V} \quad (3.5)$$

- The Second Law implies that energy flows spontaneously from an object with a shallow S - U slope (high T) to one with a steep S - U slope (low T) [p. 5885-5888].
- For a "normal" system, U increases with T , and the S - U graph is concave-down.
- For a monatomic ideal gas, this definition correctly yields $U = \frac{3}{2}NkT$ [p. 6114-6115], verifying the equipartition theorem.

3.2 Entropy and Heat

This section (partially summarized) connects the statistical definition of entropy to measurable quantities like heat capacity.

- **Predicting Heat Capacities:** We can predict C_V from a microscopic model using a 5-step procedure:
 1. Find multiplicity Ω as a function of U, V, N .
 2. Find entropy: $S = k \ln \Omega$.
 3. Find temperature: $1/T = (\partial S / \partial U)_{N,V}$.
 4. Solve for energy $U(T)$.
 5. Find heat capacity: $C_V = (\partial U / \partial T)_V$.
- For an Einstein solid (high T) or monatomic ideal gas, this procedure correctly gives $C_V = \frac{f}{2}Nk$ [p. 5988-5991].
- **Third Law of Thermodynamics:** At $T = 0$, a system settles into its unique lowest-energy state (ground state), so $\Omega = 1$ and $S = 0$ [p. 6100]. This implies that $C_V \rightarrow 0$ as $T \rightarrow 0$ [p. 6107].

3.3 Paramagnetism

This section applies these statistical methods to a two-state paramagnet.

- **Model:** N spin-1/2 dipoles in a magnetic field B . Each dipole has two states: "up"($\uparrow$) with energy $-\mu B$, or "down"($\downarrow$) with energy $+\mu B$ [p. 6157-6158].
- The total energy U and magnetization M are:

$$U = N_{\downarrow}(\mu B) + N_{\uparrow}(-\mu B) = (N_{\downarrow} - N_{\uparrow})\mu B \quad (3.25)$$

$$M = (N_{\uparrow} - N_{\downarrow})\mu = -U/B \quad (3.26)$$

- **Multiplicity:** The number of ways to arrange $N_\uparrow$ up-dipoles and $N_\downarrow$ down-dipoles.

$$\Omega(N_\uparrow) = \binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow! N_\downarrow!} \quad (3.27)$$

- **Entropy and Temperature:** Using Stirling's approximation ($\ln N! \approx N \ln N - N$ [p. 5911]) on $S = k \ln \Omega$ gives:

$$S/k \approx N \ln N - N_\uparrow \ln N_\uparrow - N_\downarrow \ln N_\downarrow \quad (3.28)$$

Applying $1/T = \partial S / \partial U$ gives the temperature:

$$\frac{1}{T} = \frac{k}{2\mu B} \ln \left(\frac{N_\downarrow}{N_\uparrow} \right) = \frac{k}{2\mu B} \ln \left(\frac{N - U/(\mu B)}{N + U/(\mu B)} \right) \quad (3.30)$$

- **Negative Temperature:** This system has a finite maximum energy (all spins $\downarrow$). When $U > 0$ (more spins $\downarrow$), the S - U graph has a negative slope, resulting in a *negative* temperature. Negative T is "hotter" than positive T .
- **Energy and Magnetization (Analytic Solution):** Solving Eq. 3.30 for U and M :

$$U = -N\mu B \tanh \left(\frac{\mu B}{kT} \right) \quad (3.31)$$

$$M = N\mu \tanh \left(\frac{\mu B}{kT} \right) \quad (3.32)$$

At high T ($kT \gg \mu B$), $\tanh(x) \approx x$, and $M \approx N\mu^2 B / (kT)$. This $1/T$ dependence is **Curie's Law**[p. 6183].

- **Heat Capacity:** Differentiating $U(T)$ gives C_V :

$$C_V = \left(\frac{\partial U}{\partial T} \right)_B = Nk \left(\frac{\mu B}{kT} \right)^2 \frac{1}{\cosh^2(\mu B/kT)} \quad (3.33)$$

This function approaches zero at both low and high T , peaking when $kT \approx \mu B$.

3.4 Mechanical Equilibrium and Pressure

This section defines pressure based on entropy.

- Consider two systems that can exchange both energy and volume. At equilibrium, S_{total} is maximum, which requires $(\partial S_A / \partial U_A) = (\partial S_B / \partial U_B)$ (i.e., $T_A = T_B$) and also:

$$\left(\frac{\partial S_A}{\partial V_A} \right)_{U,N} = \left(\frac{\partial S_B}{\partial V_B} \right)_{U,N} \quad (3.38)$$

- **Pressure (P):** This common quantity, which governs volume exchange, is defined as:

$$P \equiv T \left(\frac{\partial S}{\partial V} \right)_{U,N} \quad (3.39)$$

A system with a large $\partial S/\partial V$ "wants" to expand, so it has high pressure.

- For a monatomic ideal gas, plugging the Sackur-Tetrode equation (2.49) into this definition correctly yields the ideal gas law: $P = NkT/V$.
- **Thermodynamic Identity (for U):** This fundamental relation combines the definitions of T and P . If U is considered a function of S and V (for fixed N), the total differential is:

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Using the definitions of T and P , this becomes:

$$dU = TdS - PdV \quad (3.46)$$

This relates changes in U , S , and V for any infinitesimal, quasistatic process.

3.5 Diffusive Equilibrium and Chemical Potential

This section defines the chemical potential, which governs the exchange of particles.

- Consider two systems that can exchange energy and particles. At equilibrium, $T_A = T_B$ and $P_A = P_B$. There is a third condition:

$$\left(\frac{\partial S_A}{\partial N_A} \right)_{U,V} = \left(\frac{\partial S_B}{\partial N_B} \right)_{U,V} \quad (3.53)$$

- **Chemical Potential (μ):** This quantity is defined as:

$$\mu \equiv -T \left(\frac{\partial S}{\partial N} \right)_{U,V} \quad (3.55)$$

At equilibrium, $\mu_A = \mu_B$. Particles tend to flow spontaneously from a system of high μ to a system of low μ .

- **Generalized Thermodynamic Identity:** The thermodynamic identity (3.46) is expanded to include changes in N :

$$dU = TdS - PdV + \mu dN \quad (3.58)$$

- This identity provides alternative formulas for T , P , and μ :

$$T = \left(\frac{\partial U}{\partial S} \right)_{V,N} \quad P = - \left(\frac{\partial U}{\partial V} \right)_{S,N} \quad \mu = \left(\frac{\partial U}{\partial N} \right)_{S,V} \quad (3.60)$$

So, μ is the energy required to add one particle while keeping S and V constant.

- **Chemical Potential of an Ideal Gas:** Using the Sackur-Tetrode equation (2.49), we find μ for a monatomic ideal gas:

$$\mu = -kT \ln \left[\frac{V}{N} \left(\frac{2\pi m kT}{h^2} \right)^{3/2} \right] \quad (3.63)$$

Summary and a Look Ahead

This section summarizes the three types of equilibrium and the thermodynamic identity.

- **Thermal Equilibrium:** Governed by $T = (\partial U / \partial S)_{V,N}$.
- **Mechanical Equilibrium:** Governed by $P = -(\partial U / \partial V)_{S,N}$.
- **Diffusive Equilibrium:** Governed by $\mu = (\partial U / \partial N)_{S,V}$.
- The **thermodynamic identity** combines all three: $dU = TdS - PdV + \mu dN$.

5 Free Energy and Chemical Thermodynamics

5.1 Free Energy as Available Work

This section introduces thermodynamic potentials useful for non-isolated systems.

- **Enthalpy (H):** The total energy required to create a system at constant pressure P .

$$H \equiv U + PV \quad (5.1)$$

- **Helmholtz Free Energy (F):** The energy that must be provided as work to create a system at constant temperature T . TS is the heat provided "for free" by the environment.

$$F \equiv U - TS \quad (5.2)$$

- **Gibbs Free Energy (G):** The "other" work (non- PV work) required to create a system in an environment at constant T and P .

$$G \equiv U - TS + PV \quad (5.3)$$

These four quantities (U, H, F, G) are called **thermodynamic potentials**.

- **Changes in Free Energy:** For any process at constant T and P :

$$\Delta G \leq W_{\text{other}} \quad (5.8)$$

The change in G is the minimum "other" (e.g., electrical) work required (or maximum work that can be extracted).

- **Thermodynamic Identities:** These summarize the relationships between the potentials and their natural variables (S, T, P, V).

$$dU = TdS - PdV + \mu dN \quad (5.16)$$

$$dH = TdS + VdP + \mu dN \quad (5.18)$$

$$dF = -SdT - PdV + \mu dN \quad (5.20)$$

$$dG = -SdT + VdP + \mu dN \quad (5.23)$$

- **Partial Derivative Relations:** From the identities, we can find key variables by differentiation. From the dG identity:

$$S = -\left(\frac{\partial G}{\partial T}\right)_{P,N} \quad V = \left(\frac{\partial G}{\partial P}\right)_{T,N} \quad \mu = \left(\frac{\partial G}{\partial N}\right)_{T,P} \quad (5.24)$$

5.2 Free Energy as a Force toward Equilibrium

This section establishes the equilibrium conditions for systems in contact with an environment.

- The Second Law (S_{total} tends to increase) implies that for a system in contact with a reservoir:
 - At constant T and V , **Helmholtz Free Energy (F) tends to decrease.**
 - At constant T and P , **Gibbs Free Energy (G) tends to decrease.**
- $F = U - TS$ represents the competition between minimizing energy (U) and maximizing entropy (S). At low T , the U term dominates; at high T , the S term dominates.
- **Extensive vs. Intensive:**
 - **Extensive** quantities double when the system size doubles (V, N, S, U, H, F, G).
 - **Intensive** quantities are unchanged (T, P, μ , density).
- **Gibbs Free Energy and Chemical Potential:** Because G is extensive and T and P are intensive, G must be directly proportional to N . The constant of proportionality is μ .

$$G = N\mu \quad (5.35)$$

Therefore, μ is simply the Gibbs free energy per particle.

- **Chemical Potential of an Ideal Gas:** Using $V = (\partial G / \partial P)_{T,N} = N(\partial \mu / \partial P)_{T,N}$ and the ideal gas law ($V = NkT/P$), we can integrate to find $\mu(P)$:

$$\mu(T, P) = \mu^\circ(T) + kT \ln(P/P^\circ) \quad (5.40)$$

where P° is a standard reference pressure (e.g., 1 bar) and $\mu^\circ(T)$ is the chemical potential at that pressure.

6 Boltzmann Statistics

Calculating $\Omega(U)$ directly is often too difficult. This chapter introduces a more powerful method for systems in equilibrium with a reservoir at a fixed temperature T , rather than systems with fixed energy U .

6.1 The Boltzmann Factor

This section derives the probability of a system being in a specific microstate.

- Consider a small system (S) in thermal contact with a large reservoir (R). The total energy is fixed.
- The probability of the system being in a particular microstate s_1 is proportional to the multiplicity of the reservoir in that state: $P(s_1) \propto \Omega_R(s_1)$.
- The ratio of probabilities for two different microstates s_1 and s_2 is:

$$\frac{P(s_2)}{P(s_1)} = \frac{\Omega_R(s_2)}{\Omega_R(s_1)} = e^{(S_R(s_2) - S_R(s_1))/k}$$

- Using $dS_R = dU_R/T = -dE_S/T$, this becomes:

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{-E(s_2)/kT}}{e^{-E(s_1)/kT}} \quad (6.5)$$

- **Boltzmann Factor:** The term $e^{-E(s)/kT}$ is the Boltzmann factor. It gives the relative probability of a microstate.
- **Probability of a Microstate s :**

$$P(s) = \frac{1}{Z} e^{-E(s)/kT} \quad (6.8)$$

- **Partition Function (Z):** Z is the normalization constant, ensuring all probabilities sum to 1.

$$Z \equiv \sum_s e^{-E(s)/kT} \quad (6.10)$$

The sum is over all possible microstates s of the system.

6.2 Average Values

This section shows how to use the partition function to find average properties.

- The average value of any quantity X is found by summing $X(s)$ for each microstate, weighted by its probability $P(s)$:

$$\bar{X} = \sum_s X(s)P(s) = \frac{1}{Z} \sum_s X(s)e^{-E(s)/kT} \quad (6.18)$$

- **Average Energy ($\bar{E}$ or U):** The average energy of a system is:

$$\bar{E} = \frac{1}{Z} \sum_s E(s) e^{-E(s)/kT}$$

This can be expressed concisely as a derivative of Z . Using $\beta = 1/kT$:

$$\bar{E} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial(\ln Z)}{\partial \beta} \quad (\text{Prob 6.16})$$

- **Example: Two-State Paramagnet.** For a single dipole, $Z = e^{\beta\mu B} + e^{-\beta\mu B} = 2 \cosh(\beta\mu B)$.
- The average energy is:

$$\bar{E} = -\frac{\partial}{\partial \beta} \ln[2 \cosh(\beta\mu B)] = -\mu B \tanh(\beta\mu B) \quad (6.23)$$

- The total energy for N dipoles is $U = N\bar{E}$, matching Eq. (3.31).

6.3 The Equipartition Theorem

This section provides a formal proof of the equipartition theorem.

- This theorem applies only to degrees of freedom that are **quadratic** ($E(q) = cq^2$) and can be treated **classically** (energy levels are very closely spaced compared to kT).
- The partition function for one such degree of freedom (treating the variable q as continuous) is:

$$Z = \int_{-\infty}^{\infty} e^{-\beta cq^2} dq' \propto \int_{-\infty}^{\infty} e^{-x^2} dx \propto \beta^{-1/2}$$

(where dq' is a small unit of q to make Z dimensionless, and $x^2 = \beta cq^2$).

- We can write $Z = C\beta^{-1/2}$, where C is some constant.
- The average energy for this one degree of freedom is:

$$\bar{E} = -\frac{\partial(\ln Z)}{\partial \beta} = -\frac{\partial}{\partial \beta} (\ln C - \frac{1}{2} \ln \beta) = -\left(0 - \frac{1}{2} \frac{1}{\beta}\right) = \frac{1}{2\beta}$$

- Since $\beta = 1/kT$, we have the equipartition theorem:

$$\bar{E} = \frac{1}{2} kT \quad (6.40)$$

6.4 The Maxwell Speed Distribution

This section derives the probability distribution for the speeds of molecules in an ideal gas.

- The probability of a molecule having a speed in the range v to $v + dv$ is $D(v)dv$, where $D(v)$ is the Maxwell speed distribution [p. 6395-6397].
- $D(v)$ is proportional to two factors:
 1. The Boltzmann factor for translational kinetic energy: $e^{-mv^2/2kT}$ [p. 6389-6391].
 2. The number of velocity vectors corresponding to that speed. This is proportional to the surface area of a sphere in "velocity space" with radius v : $4\pi v^2$ [p. 6393].
- **Maxwell Speed Distribution:** The normalized probability distribution is:

$$D(v) = \left(\frac{m}{2\pi kT} \right)^{3/2} 4\pi v^2 e^{-mv^2/2kT} \quad (6.50)$$

- From this distribution, characteristic speeds can be calculated:
- **Most Probable Speed ($v_{\max}$):** The speed where $D(v)$ is maximum.

$$v_{\max} = \sqrt{\frac{2kT}{m}}$$

- **Average Speed ($\bar{v}$):** The mean speed of all molecules.

$$\bar{v} = \int_0^\infty v D(v) dv = \sqrt{\frac{8kT}{\pi m}} \quad (6.52)$$

- **RMS Speed (v_{rms}):** The square root of the average squared speed.

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}} \quad (6.41)$$

6.5 Partition Functions and Free Energy

This section connects the microscopic partition function (Z) to the macroscopic Helmholtz free energy (F).

- **The Fundamental Relation:** The Helmholtz free energy F of a system at temperature T is given by its partition function Z :

$$F = -kT \ln Z \quad (6.56)$$

This is a crucial bridge from the statistical (microscopic) Z to the thermodynamic (macroscopic) F .

- **Proof Sketch:** This relation is proven by showing that $F' = -kT \ln Z$ obeys the same differential equation as $F = U - TS$, namely $(\partial F / \partial T) = (F - U) / T$ (Eq. 6.59), and has the same value at $T = 0$ ($F(0) = U_0$).
- **Calculating All Thermodynamic Properties from Z :** Once F is known from Z , all other thermodynamic quantities can be found using the partial derivative relations from the thermodynamic identity for F :

$$\begin{aligned} S &= - \left(\frac{\partial F}{\partial T} \right)_{V,N} = k \ln Z + kT \left(\frac{\partial(\ln Z)}{\partial T} \right)_{V,N} \\ P &= - \left(\frac{\partial F}{\partial V} \right)_{T,N} = kT \left(\frac{\partial(\ln Z)}{\partial V} \right)_{T,N} \end{aligned} \quad (6.64)$$

$$\mu = - \left(\frac{\partial F}{\partial N} \right)_{T,V} = -kT \left(\frac{\partial(\ln Z)}{\partial N} \right)_{T,V} \quad (6.64)$$

6.6 Partition Functions for Composite Systems

This section relates the total partition function of a system (Z_{total}) to the partition functions of its individual components (e.g., Z_1 for a single particle).

- **Noninteracting, Distinguishable Particles:** The total partition function is the product of the individual partition functions.

$$Z_{\text{total}} = Z_1 Z_2 Z_3 \cdots Z_N = (Z_1)^N \quad (6.69)$$

- **Noninteracting, Indistinguishable Particles (Low-Density Limit):**
- For indistinguishable particles (e.g., identical molecules in a gas), we must correct for overcounting states by dividing by $N!$.

$$Z_{\text{total}} = \frac{1}{N!} (Z_1)^N \quad (6.70)$$

- This approximation is valid when the number of available single-particle states is much larger than the number of particles ($Z_1 \gg N$), so the probability of two particles occupying the same state is negligible [p. 6432, 6434].

6.7 Ideal Gas Revisited

This section uses the partition function method to re-derive all thermodynamic properties of an ideal gas.

- The partition function for a single molecule (Z_1) is factored into translational and internal parts:

$$Z_1 = Z_{\text{trans}} Z_{\text{int}} \quad (6.73)$$

- **Translational Partition Function (Z_{trans}):** Found by summing over the quantized energy states of a particle in a 3D box ($E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$) [p. 6458, 6461-6462]. This sum is approximated by an integral:

$$Z_{\text{trans}} = \sum_n e^{-E_n/kT} \approx \int e^{-E_n/kT} d^3n = \frac{V}{v_Q} \quad (6.79)$$

- **Quantum Volume (v_Q):** The typical volume of a particle's wave packet.

$$v_Q = \left(\frac{h^2}{2\pi m k T} \right)^{3/2} \quad (6.83)$$

Z_{trans} is the number of quantum volumes that fit inside the box V .

- **Total Partition Function (Z):** For N indistinguishable molecules:

$$Z = \frac{1}{N!} (Z_1)^N = \frac{1}{N!} \left(\frac{V Z_{\text{int}}}{v_Q} \right)^N \quad (6.85)$$

- **Thermodynamic Properties from Z :**
- **Helmholtz Free Energy (F):** Using $F = -kT \ln Z$ (Eq. 6.56) and Stirling's approximation ($\ln N! \approx N \ln N - N$):

$$F = -NkT \left[\ln \left(\frac{V Z_{\text{int}}}{N v_Q} \right) + 1 \right] \quad (6.90)$$

- **Pressure (P):**

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T,N} = -(-NkT/V) = \frac{NkT}{V} \quad (6.91)$$

This successfully derives the ideal gas law from quantum statistics.

- **Entropy (S):** $S = -(\partial F / \partial T)_{V,N}$. For a monatomic gas ($Z_{\text{int}} = 1$), this yields the **Sackur-Tetrode Equation**:

$$S = Nk \left[\ln \left(\frac{V}{N v_Q} \right) + \frac{5}{2} \right] \quad (6.92)$$

- **Energy (U):** $U = -\partial(\ln Z) / \partial \beta$.

$$U = \frac{3}{2} NkT + N \bar{E}_{\text{int}} \quad (6.88)$$

The energy is the sum of translational (from equipartition) and internal (rotational, vibrational) parts.

- **Chemical Potential:**

$$\mu = -kT \ln \left(\frac{V Z_{\text{int}}}{N v_Q} \right), \quad (6.93)$$

$$v_Q = \left(\frac{h}{\sqrt{2\pi m k T}} \right)^3.$$

For monatomic gas $Z_{\text{int}} = 1$.

7 Quantum Statistics

7.1 The Gibbs Factor

This section introduces statistical mechanics for a system that can exchange both energy and particles with a reservoir at constant T and μ .

- **Gibbs Factor:** The relative probability of a microstate s with energy $E(s)$ and $N(s)$ particles.

$$\text{Gibbs factor} = e^{-[E(s)-\mu N(s)]/kT} \quad (7.5)$$

- **Probability of a Microstate s :**

$$P(s) = \frac{1}{\mathcal{Z}} e^{-[E(s)-\mu N(s)]/kT} \quad (7.6)$$

- **Grand Partition Function ($\mathcal{Z}$):** The normalization constant, also called the Gibbs sum for fermions.

$$\mathcal{Z} = \sum_s e^{-[E(s)-\mu N(s)]/kT} \quad (7.7)$$

The sum is over all possible microstates, including all possible values of $N(s)$. For **Bosons** it looks like

$$\mathcal{Z} = \prod_k \left(\sum_{n=0}^{\infty} e^{-\frac{n(E_k-\mu)}{k_b T}} \right).$$

- **Example: Adsorption Site.** A site (system) can be empty ($E=0, N=0$) or occupied ($E=\epsilon, N=1$). The grand partition function is $\mathcal{Z} = 1 + e^{-(\epsilon-\mu)/kT}$ [p. 6411]. The probability of the site being occupied is:

$$P(\text{occupied}) = \frac{e^{-(\epsilon-\mu)/kT}}{1 + e^{-(\epsilon-\mu)/kT}}$$

- **Average Number of Particles in a System:** When using the Grand Partition Function:

$$\bar{N} = \frac{k_b T}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \mu}$$

7.2 Bosons and Fermions

This section introduces the two fundamental types of quantum particles and their distribution functions.

- **Bosons:** Particles with integer spin (e.g., photons, ^4He). Any number of identical bosons can occupy the same single-particle state.
- **Fermions:** Particles with half-integer spin (e.g., electrons, protons, ^3He).

- **Pauli Exclusion Principle:** Two identical fermions cannot occupy the same single-particle state.
- **Classical Limit ($Z_1 \gg N$):** When the number of available single-particle states (Z_1) is much greater than the number of particles (N), the probability of two particles occupying the same state is negligible. In this limit ($V/N \gg v_Q$), both quantum distributions reduce to the classical Boltzmann distribution.
- **Distribution Functions:** We find the average number of particles ($\bar{n}$) in a single-particle state with energy ϵ by treating that state as the system.
- **Fermi-Dirac Distribution (Fermions):** n can be 0 or 1.

$$\bar{n}_{\text{FD}} = \frac{1}{e^{(\epsilon-\mu)/kT} + 1} \quad (7.23)$$

$\bar{n}_{\text{FD}}$ is always between 0 and 1.

- **Bose-Einstein Distribution (Bosons):** n can be 0, 1, 2, ...

$$\bar{n}_{\text{BE}} = \frac{1}{e^{(\epsilon-\mu)/kT} - 1} \quad (7.28)$$

This requires $\mu < \epsilon$ for all states. $\bar{n}_{\text{BE}}$ can be greater than 1.

- **Boltzmann Distribution (Classical Limit):**

$$\bar{n}_{\text{Boltz}} = e^{-(\epsilon-\mu)/kT} \quad (7.31)$$

This is the limit of both $\bar{n}_{\text{FD}}$ and $\bar{n}_{\text{BE}}$ when $(\epsilon - \mu) \gg kT$.

7.3 Degenerate Fermi Gases

This section applies Fermi-Dirac statistics to a gas of fermions at very low temperatures (a "degenerate" gas), such as the conduction electrons in a metal.

- **Zero Temperature ($T = 0$):** The Fermi-Dirac distribution is a step function. All states with energy $\epsilon < \mu$ are occupied ($\bar{n} = 1$), and all states with $\epsilon > \mu$ are empty ($\bar{n} = 0$).
- **Fermi Energy (ϵ_F):** The chemical potential at $T = 0$. It is the energy of the highest occupied state.

$$\epsilon_F \equiv \mu(T = 0) \quad (7.33)$$

- **Free Electron Gas:** For N non-interacting electrons in a 3D box of volume V , the allowed energies are $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$ [p. 6425].
- At $T = 0$, the occupied states fill a sphere in "n-space". The radius of this sphere, n_{max} , is related to N , which gives the Fermi energy:

$$\epsilon_F = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} \quad (7.39)$$

- The total energy at $T = 0$ is found by integrating the energy of all occupied states:

$$U = \frac{3}{5}N\epsilon_F \quad (7.42)$$

- **Fermi Temperature (T_F):** $T_F \equiv \epsilon_F/k$. For electrons in a metal, T_F is typically thousands of Kelvin, so room temperature is "very low".
- **Degeneracy Pressure:** The pressure exerted by a Fermi gas at $T = 0$ due to the Pauli exclusion principle.

$$P = -\left(\frac{\partial U}{\partial V}\right)_{S,N} = \frac{2U}{3V} = \frac{2}{5}N\frac{\epsilon_F}{V} \quad (7.44)$$

- **Small Nonzero Temperatures ($kT \ll \epsilon_F$):**
- Only electrons within $\sim kT$ of the Fermi energy can be thermally excited.
- The total energy is given by the **Sommerfeld expansion**:

$$U \approx \frac{3}{5}N\epsilon_F + \frac{\pi^2}{4}N\frac{(kT)^2}{\epsilon_F} \quad (7.47)$$

- **Heat Capacity (C_V):** Differentiating U with respect to T gives the electronic heat capacity:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\pi^2 N k^2}{2\epsilon_F} T = \frac{\pi^2}{3} g(\epsilon_F) k^2 T \quad (7.48)$$

C_V is linear in T , unlike the T^3 dependence for phonons (lattice vibrations).

- **Density of States ($g(\epsilon)$):** The number of single-particle states per unit energy. For a 3D free electron gas:

$$g(\epsilon) = \frac{3N}{2\epsilon_F^{3/2}} \sqrt{\epsilon} \quad (7.51)$$

- At any temperature T , the total number of particles N and total energy U are given by integrals over all states:

$$N = \int_0^\infty g(\epsilon) \bar{n}_{\text{FD}}(\epsilon) d\epsilon \quad (7.53)$$

$$U = \int_0^\infty \epsilon g(\epsilon) \bar{n}_{\text{FD}}(\epsilon) d\epsilon \quad (7.54)$$

7.4 Blackbody Radiation

This section treats electromagnetic radiation (photons) as a gas of bosons.

- **Ultraviolet Catastrophe:** Classical equipartition theorem predicts infinite energy in the electromagnetic field in a box ($U = \infty$) [p. 5863-5864].
- **Quantum Solution (Planck):** Energy in an EM mode (an oscillator) is quantized: $E_n = nhf$, where n is the number of energy units (photons) [p. 5873, 5883].
- **Photons:** Photons are bosons. Since they can be created and destroyed (e.g., absorbed by a wall), their total number is not conserved, which implies their **chemical potential is zero** [p. 5880-5881].
- **Planck Distribution:** Setting $\mu = 0$ and $\epsilon = hf$ in the Bose-Einstein distribution (Eq. 7.28) gives the average number of photons in a single mode of frequency f :

$$\bar{n}_{\text{Planck}} = \frac{1}{e^{hf/kT} - 1} \quad (7.72)$$

- **Density of States & Energy Density:** Summing the energy $hf \cdot \bar{n}_{\text{Planck}}$ over all allowed modes (3D box, 2 polarizations) yields the total energy U [p. 5895-5896]. The energy per unit volume, per unit energy $\epsilon = hf$, is the **Planck spectrum**:

$$u(\epsilon) = \frac{8\pi}{(hc)^3} \frac{\epsilon^3}{e^{\epsilon/kT} - 1} \quad (7.84)$$

- **Total Energy (Stefan-Boltzmann Law):** Integrating the energy density $u(\epsilon)$ over all energies ϵ gives the total energy per unit volume:

$$\frac{U}{V} = \frac{8\pi^5 (kT)^4}{15(hc)^3} \propto T^4 \quad (7.86)$$

- **Entropy of a Photon Gas:** From $C_V = (\partial U / \partial T)_V$ and $S = \int (C_V / T) dT$:

$$S = \frac{32\pi^5 V k^4}{45(hc)^3} T^3 \quad (7.89)$$

- **Blackbody Radiation:** A hole in a cavity or a perfect blackbody emits radiation. The power emitted per unit area is given by **Stefan's Law**:

$$\frac{\text{Power}}{\text{Area}} = \frac{c}{4} \frac{U}{V} = \sigma T^4 \quad (7.97)$$

where σ is the **Stefan-Boltzmann constant**, $\sigma = \frac{2\pi^5 k^4}{15h^3 c^2} = 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$ [p. 5940-5941].

- **Emissivity (e):** For a non-black object, the power radiated is reduced by a factor e (which can depend on wavelength), where $e = 1$ for a perfect blackbody and $e = 0$ for a perfect reflector [p. 5945-5946].

$$\text{Power} = e\sigma AT^4 \quad (7.99)$$

7.5 Debye Theory of Solids

This section provides a more accurate model for the heat capacity of solids, treating vibrations as quantized sound waves (phonons).

- **Problem with Einstein Model:** Predicts $C_V \propto e^{-\epsilon/kT}$ at low T , but experiments show $C_V \propto T^3$ [p. 5951, 5953]. The Einstein model fails because it assumes all oscillators have the same frequency.
- **Debye Model:** Treats lattice vibrations as **phonons** (bosons with $\mu = 0$) in a continuous medium[p. 5952, 5959].
- This model differs from the photon gas in three ways[p. 5952]:
 1. Phonons travel at the speed of sound c_s , not c .
 2. Phonons have three polarizations (1 longitudinal, 2 transverse).
 3. The total number of modes is finite ($3N$), not infinite. The wavelength cannot be shorter than the atomic spacing.
- **Debye Approximation:** The sum over all $3N$ modes is approximated by an integral over a sphere in n -space, cut off at a maximum radius $n_{\max}$ such that the total number of modes is $3N$ [p. 5956-5957].
- **Debye Temperature (T_D):** The cutoff corresponds to a maximum energy $\epsilon_D = kT_D$. T_D is the **Debye temperature**.

$$kT_D = \frac{hc_s}{2L} n_{\max} = \frac{hc_s}{2L} \left(\frac{6N}{\pi} \right)^{1/3} \quad (7.111)$$

- **Total Energy (U):**

$$U = 9NkT \left(\frac{T}{T_D} \right)^3 \int_0^{T_D/T} \frac{x^3}{e^x - 1} dx \quad (7.112)$$

- **High-Temperature Limit ($T \gg T_D$):** The integral simplifies, and $U \approx 3NkT$. This gives $C_V = 3Nk$, the classical **Dulong-Petit law**[p. 5967].
- **Low-Temperature Limit ($T \ll T_D$):** The integral's upper limit goes to ∞ , yielding $\pi^4/15$.

$$U = \frac{3\pi^4 NkT^4}{5T_D^3} \quad (7.114)$$

This gives the **Debye T^3 Law** for heat capacity, which matches experiments:

$$C_V = \frac{12\pi^4 Nk}{5} \left(\frac{T}{T_D} \right)^3 \quad (7.115)$$

- For metals at low T , C_V includes both the electronic (linear in T) and phonon (T^3) terms[p. 5970]:

$$C_V = \gamma T + AT^3 \quad (7.116)$$

7.6 Bose-Einstein Condensation

This section considers a gas of non-interacting bosons (like ^4He) where the total number of particles N is conserved.

- At $T = 0$, all N particles must occupy the ground state (lowest single-particle state ϵ_0) [p. 5975].
- **Ground State Occupancy:** The average number of particles in the ground state is $\bar{n}_0$.

$$\bar{n}_0 = \frac{1}{e^{(\epsilon_0 - \mu)/kT} - 1} \quad (7.119)$$

For $\bar{n}_0$ to be large (e.g., 10^{20}), the denominator must be tiny, which implies μ must be extremely close to ϵ_0 :

$$\mu \approx \epsilon_0 - \frac{kT}{\bar{n}_0} \quad (7.120)$$

- **Number of Excited Particles (N_{excited}):** The total N is fixed: $N = \bar{n}_0 + N_{\text{excited}}$. We can find N_{excited} by integrating the Bose-Einstein distribution over all excited states, approximating $\mu \approx \epsilon_0 = 0$.

$$N_{\text{excited}} = \int_0^\infty g(\epsilon) \frac{1}{e^{\epsilon/kT} - 1} d\epsilon \quad (7.122)$$

- **Condensation Temperature (T_c):** The integral for N_{excited} has a maximum value. The temperature at which $N_{\text{excited}} = N$ is the **condensation temperature** T_c .

$$N = 2.612 \left(\frac{2\pi m k T_c}{h^2} \right)^{3/2} V \quad \text{or} \quad kT_c = \frac{h^2}{2\pi m} \left(\frac{N}{2.612V} \right)^{2/3} \quad (7.126)$$

This is the temperature at which the quantum volume v_Q is comparable to the average volume per particle V/N .

- **Bose-Einstein Condensation (BEC):**

- For $T > T_c$: $\mu < 0$. All N particles are in excited states ($N_{\text{excited}} = N$, $\bar{n}_0 \approx 0$).
- For $T < T_c$: $\mu \approx 0$. The excited states can only hold a maximum of N_{excited} particles.

$$N_{\text{excited}} = N \left(\frac{T}{T_c} \right)^{3/2} \quad (7.128)$$

- The remaining particles, $\bar{n}_0$, must "condense" into the ground state.
- **Condensate Fraction ($\bar{n}_0$):**

$$\bar{n}_0 = N - N_{\text{excited}} = N \left[1 - \left(\frac{T}{T_c} \right)^{3/2} \right] \quad (7.129)$$

- **Real-World Examples:** BEC was first observed in ultracold atomic gases (^{87}Rb) in 1995 [p. 6003, 6005]. It is also the mechanism behind superfluidity in liquid ^4He and superconductivity (where electron pairs act as bosons) [p. 6007-6010].